G-PRINCIPAL AND G-TORSORS STACKS ARE NOT EQUIVALENT IN GENERAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is the one-line conclusion of the section's explicit Q-example; the author's omitted classification proof is not load-bearing for this card, content verbatim.

Editorial note. The label lemma-torsors-principal-spaces-not-equal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 4040-4096: G = (mu_{2,k})^infty = lim (mu_{2,k})^n, an affine group scheme with k[t_1, t_2, ...]/(t_i^2 - 1); the torsor P = Spec(k[x_1, x_2, ...]/(x_i^2 - a_i)) with G-action x_i -> t_i tensor x_i, which is a G-principal space (over k' = k(sqrt a_1, ...) it becomes G-equivariantly isomorphic to G) and is an fppf torsor if and only if k'/k is finite (forward via the fppf covering Spec(k'), backward via a closed point of a trivilizing cover and Varieties, lemma-locally-finite-type-Jacobson); explicit example k = Q, a_i = i gives k'/k infinite
- Examples of Stacks subsections defining G-Principal and G-Torsors

Source reference. Original reference 04UR: sources/stacks-chow/tags.txt, source lines 6299–6299. Prerequisite pointer invoked at 4105; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04UN: sources/stacks-chow/tags.txt, source lines 6296–6296. Prerequisite pointer invoked at 4102; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 040M: sources/stacks-chow/tags.txt, source lines 5261–5261. Prerequisite pointer invoked at 4130; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 049C: sources/stacks-chow/tags.txt, source lines 5575–5575. Prerequisite pointer invoked at 4116; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0478: sources/stacks-chow/tags.txt, source lines 5499–5499. Prerequisite pointer invoked at 4162; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

In Groupoids, Remark 049C we raise the question whether any G-torsor is a G-torsor for the fppf topology. In this section we show that this is not always the case.

Let k be a field. All schemes and stacks are over k in what follows. Let $G \to \mathrm{Spec}(k)$ be the group scheme

$$G = (\mu_{2,k})^\infty = \mu_{2,k} \times_k \mu_{2,k} \times_k \mu_{2,k} \times_k \ldots = \lim_n (\mu_{2,k})^n$$

where $\mu_{2,k}$ is the group scheme of second roots of unity over $\mathrm{Spec}(k)$, see Groupoids, Example 040M. As an inverse limit of affine schemes we see that G is an affine group scheme. In fact it is the spectrum of the ring $k[t_1, t_2, t_3, \ldots]/(t_i^2 - 1)$. The multiplication map $m : G \times_k G \to G$ is on the algebra level given by $t_i \mapsto t_i \otimes t_i$.

We claim that any G-torsor over k is of the form

$$P = \mathrm{Spec}(k[x_1, x_2, x_3, \ldots]/(x_i^2 - a_i))$$

for certain $a_i \in k^*$ and with G-action $G \times_k P \to P$ given by $x_i \to t_i \otimes x_i$ on the algebra level. We omit the proof. Actually for the example we only need that P is a G-torsor which is clear since over $k' = k(\sqrt{a_1}, \sqrt{a_2}, \ldots)$ the scheme P becomes isomorphic to G in a G-equivariant manner. Note that P is trivial if and only if $k' = k$ since if P has a k-rational point then all of the a_i are squares.

We claim that P is an fppf torsor if and only if the field extension $k' = k(\sqrt{a_1}, \sqrt{a_2}, \ldots)/k$ is finite. If k' is finite over k, then $\{\mathrm{Spec}(k') \to \mathrm{Spec}(k)\}$ is an fppf covering which trivializes P and we see that P is indeed an fppf torsor. Conversely, suppose that P is an fppf G-torsor. This means that there exists an fppf covering $\{S_i \to \mathrm{Spec}(k)\}$ such that each P_{S_i} is trivial. Pick an i such that S_i is not empty. Let $s \in S_i$ be a closed point. By Varieties, Lemma 0478 the field extension $\kappa(s)/k$ is finite, and by construction $P_{\kappa(s)}$ has a $\kappa(s)$-rational point. Thus we see that $k \subset k' \subset \kappa(s)$ and k' is finite over k.

To get an explicit example take $k = \mathbf{Q}$ and $a_i = i$ for example (or a_i is the ith prime if you like).

Lemma 0.1. *Let S be a scheme. Let G be a group scheme over S. The stack G-Principal classifying principal homogeneous G-spaces (see Examples of Stacks, Subsection 04UN) and the stack G-Torsors classifying fppf G-torsors (see Examples of Stacks, Subsection 04UR) are not equivalent in general.*

Proof. The discussion above shows that the functor *G-Torsors* $\to$ *G-Principal* isn't essentially surjective in general. □